An explicit nonreversible Cauchy semigroup on the circle: kernel, generator and entropy

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Abstract

A fractional circle generator with sine drift admits an explicit transition kernel obtained by harmonic extension along a rational disk flow. We construct its conservative Feller semigroup and invariant wrapped Cauchy law, identify the full generator graph on continuous functions and its Laplace resolvent, and prove graph density of time-smoothed trigonometric data. For every initial probability law the positive-time forward density has finite relative entropy. Its exact entropy production is a singular jump-Bregman integral, even though the generator is nonsymmetric under its invariant law. An elementary nonsharp modified logarithmic-Sobolev bound yields exponential decay from initial time, with infinite initial entropy allowed. The proof avoids assigning an unproved backward-generator domain to a logarithmic density. The construction belongs to classical harmonic and semigroup theory and is related to known Cauchy-driven phase dynamics. Exact statements are mapped to a frozen Lean development; novelty is unresolved.

1 The circle flow and the conclusions

Let $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ and let m be normalized Haar measure, so in angle coordinates $dm = d\theta/(2\pi)$. For $a \in \mathbb{D}$, the Poisson kernel is

$$p_a(w) = \frac{1 - |a|^2}{|w - a|^2}, \qquad w \in \mathbb{T}.$$

It is a positive probability density with respect to m. Fix a real parameter $|q| < 1$, and put

$$r = \frac{2q}{1 - q^2}, \qquad \alpha = \frac{1 + q^2}{1 - q^2}, \qquad c_q = \frac{1 - |q|}{2(1 + |q|)}. \tag{1}$$

For $0 \le \lambda \le 1$ define

$$F_\lambda(z) = \frac{q(1 - \lambda) + (q^2 + \lambda)z}{1 + q^2\lambda + q(1 - \lambda)z}, \qquad \phi_t = F_{e^{-\alpha t}} \quad (t \ge 0). \tag{2}$$

For $t > 0$ define the transition probability

$$K_t(z, dw) = p_{\phi_t(z)}(w)\, m(dw), \qquad K_0(z, \cdot) = \delta_z, \tag{3}$$

and $P_t f(z) = \int f(w) K_t(z, dw)$. Our invariant reference law is $\nu_q(dw) = p_q(w) m(dw)$.

Theorem 1.1. *The kernels* (3) *form a conservative Markov semigroup. On* $C(\mathbb{T};\mathbb{C})$*,* P_t *is a strongly continuous positive contraction semigroup with invariant probability* ν_q*. Its closed generator* A *acts on trigonometric polynomials as*

$$A = -|D_\theta| - r\sin\theta\,\partial_\theta, \qquad |D_\theta|e^{in\theta} = |n|e^{in\theta}. \tag{4}$$

Its full graph is characterized by

$$(f,g)\in \operatorname{graph} A \quad\Longleftrightarrow\quad P_t f - f = \int_0^t P_s g\,ds \quad \text{for every } t\ge 0. \tag{5}$$

Time averages of trigonometric polynomials are graph dense. For $\beta > 0$ *the positive Laplace resolvent* $R_\beta f = \int_0^\infty e^{-\beta t}P_t f\,dt$ *is the inverse of* $\beta - A$*, with norm at most* $1/\beta$*.*

For every Borel probability measure μ*, write* $\mu_t = \mu K_t$*. At every* $t > 0$ *it has a positive continuous density* $f_t = d\mu_t/d\nu_q$*, and*

$$\frac{d}{dt}D(\mu_t\|\nu_q) = -\int_{\mathbb{T}^2}\frac{2\,B(f_t(z),f_t(w))}{|w-z|^2 p_q(w)}\,\nu_q(dz)\nu_q(dw), \tag{6}$$

where $B(a,b) = a\log(a/b) - a + b$ *and the diagonal integrand is defined to be zero. The integral is finite, and*

$$D(\mu_t\|\nu_q) \le e^{-c_q t}D(\mu\|\nu_q), \qquad t\ge 0, \tag{7}$$

in the extended nonnegative reals. The invariant law is unique. For $q \neq 0$ *the generator is not Hermitian even on its finite Fourier domain in* $L^2(\nu_q)$*.*

Here $D(\mu\|\nu) = \int f\log f\,d\nu$ when $\mu = f\nu$ and this integral is finite, with the usual extended-value definition otherwise. The constant c_q is a convenient lower bound, not a sharp entropy constant. The generator graph is on continuous functions with the supremum norm. We will not identify it with an unproved Sobolev domain or replace the smoothed graph core by an unsmoothed-core assertion.

2 Disk geometry and harmonic transition kernels

The denominator in (2) has modulus at least $1 + q^2\lambda - |q|(1-\lambda) > 0$ on the closed disk. Writing its numerator and denominator as N and H, direct multiplication gives

$$\begin{aligned}|H|^2 - |N|^2 = {}& (1-\lambda)^2(1-q^2)|1+qz|^2 \\ &+ \lambda^2(1+q^2)^2(1-|z|^2) \\ &+ 2\lambda(1-\lambda)(1+q^2)(1-q^2|z|^2).\end{aligned} \tag{8}$$

All terms are nonnegative on $\overline{\mathbb{D}}$, and the first is strictly positive for $\lambda < 1$. Thus F_λ maps the closed disk strictly into $\mathbb{D}$ when $\lambda < 1$, and $F_1(z) = z$. Each map is holomorphic on a neighborhood of the closed disk.

The matrix defining this fractional-linear map is

$$M_\lambda = \begin{pmatrix} q^2+\lambda & q(1-\lambda) \\ q(1-\lambda) & 1+q^2\lambda \end{pmatrix}.$$

Its eigenvectors $(q,1)$ and $(1,-q)$ have eigenvalues $1+q^2$ and $\lambda(1+q^2)$, respectively. Hence $M_\lambda M_\eta = (1+q^2)M_{\lambda\eta}$, proving

$$\phi_{s+t} = \phi_s \circ \phi_t, \qquad \phi_0(z) = z, \qquad \phi_t(q) = q. \tag{9}$$

Differentiating (2) at time zero gives

$$\partial_t \phi_t(z)|_{t=0} = V(z) := -z + \frac{q}{1-q^2}(1-z^2). \tag{10}$$

The denominator lower bound makes this differentiation uniform on the closed disk. It also gives joint continuity in t, z and uniform convergence $\phi_t(z) \to z$ as $t \downarrow 0$.

For $f \in C(\mathbb{T};\mathbb{C})$, let u_f be its harmonic Poisson extension to the disk, continuous up to the boundary. Then

$$P_t f(z) = u_f(\phi_t(z)).$$

The function $u_f \circ \phi_s$ is harmonic in $\mathbb{D}$ and continuous on its closure. Its boundary values are $P_s f$. The uniqueness of the Dirichlet problem therefore gives $u_{P_s f} = u_f \circ \phi_s$, whence

$$P_t P_s f = u_f \circ \phi_s \circ \phi_t = P_{s+t} f.$$

Equality on continuous functions determines finite Borel measures on the compact circle. This proves the exact Chapman–Kolmogorov identity for (3), not just agreement of a formal generator.

The Poisson kernel is nonnegative with integral one, so P_t is positive, preserves constants, and is a supremum-norm contraction. Continuity of $u_f \circ \phi_t$ proves the Feller property. For strong continuity at zero, use uniform continuity of u_f on $\overline{\mathbb{D}}$ and uniform convergence of ϕ_t to the identity. One can also verify boundary continuity of the Poisson extension directly: split its integral into a small arc where f oscillates little and its complement, where $p_a(w) \le (1-|a|^2)/(\varepsilon - |a-z|)^2$ tends uniformly to zero as $a \to z \in \mathbb{T}$.

Finally $\int_{\mathbb{T}} f \, d\nu_q = u_f(q)$, so

$$\int_{\mathbb{T}} P_t f \, d\nu_q = u_{P_t f}(q) = u_f(\phi_t(q)) = u_f(q).$$

This proves invariance. The kernels are Borel measurable in their initial point because their positive-time densities are jointly continuous, and at zero they are the measurable identity kernel.

3 The closed generator and the precise graph core

For $n \ge 0$, the Poisson moment identity gives

$$P_t(z^n) = \phi_t(z)^n, \qquad P_t(\bar{z}^n) = \overline{\phi_t(z)}^{\,n}. \tag{11}$$

Uniform differentiation using (10) shows that these modes belong to the continuous-function generator domain. For $n \ge 1$,

$$Az^n = -nz^n + \frac{nr}{2}(z^{n-1} - z^{n+1}),$$

with the conjugate formula for negative frequencies and $A1 = 0$. Since $\partial_\theta z^n = inz^n$, these are exactly (4). This statement concerns actual supremum-norm difference quotients.

Define $f \in D(A)$ when $h^{-1}(P_h f - f)$ has a supremum-norm limit g as $h \downarrow 0$, and set $Af = g$. The limit is unique.

Proposition 3.1. *The equivalence* (5) *holds, and its right-hand side defines a closed linear graph.*

Proof. If $Af = g$, the semigroup identity gives the right derivative $\partial_t^+ P_t f = P_t g$. For $t > 0$ the left difference quotient is $P_{t-h}(P_h f - f)/h$, also tending to $P_t g$ by contraction and strong continuity. Thus the orbit is differentiable for $t > 0$ with continuous derivative $P_t g$, and the vector-valued fundamental theorem of calculus gives (5).

Conversely, divide that identity by t and let $t \downarrow 0$. Strong continuity gives $t^{-1} \int_0^t P_s g \, ds \to g$, so $f \in D(A)$ and $Af = g$. For a convergent sequence of graph pairs (f_n, g_n), contraction bounds the difference of the integrals by $t\|g_n - g\|_\infty$. Pass to the limit in (5); the limiting pair is again in the graph. Linearity follows from the same identity. □

For $h > 0$ define the time average

$$J_h f = \frac{1}{h} \int_0^h P_s f \, ds. \tag{12}$$

A shift of the integration interval gives

$$J_h f \in D(A), \qquad A J_h f = (P_h f - f)/h. \tag{13}$$

Indeed $(P_t J_h f - J_h f)/t$ is $1/h$ times the difference of the averages of $P_s f$ on $[h, h+t]$ and $[0, t]$. Letting $t \downarrow 0$ proves the formula. Since $J_h f \to f$, the generator domain is dense. If $f \in D(A)$, (5) also gives $A J_h f = J_h A f \to Af$, so these averages approximate in the full graph norm.

Let $\mathcal{P}$ be the real-and-complex trigonometric polynomials. They are uniformly dense in $C(\mathbb{T}; \mathbb{C})$. For fixed $h > 0$, the map

$$f \longmapsto (J_h f, (P_h f - f)/h)$$

is continuous into the product of supremum-norm spaces: its two component norms are bounded by 1 and $2/h$. For a graph pair (f, Af), first choose h small enough that its averaged pair is close; then choose $p \in \mathcal{P}$ close enough to f that the corresponding averaged pair is close. Consequently

$$\overline{\{(J_h p, A J_h p) : h > 0, \ p \in \mathcal{P}\}} = \operatorname{graph} A. \tag{14}$$

This is the precise smoothed core assertion. It does not identify the closure of the unsmoothed polynomial graph.

4 Laplace resolvent on the full domain

For $\beta > 0$, contraction makes the Bochner integral

$$R_\beta f = \int_0^\infty e^{-\beta t} P_t f \, dt$$

absolutely convergent, with norm at most $\|f\|_\infty/\beta$. It is positive on real nonnegative inputs. For a finite integral $u_T = \int_0^T e^{-\beta t} P_t f \, dt$, the shifted-orbit identity is

$$P_h u_T = e^{\beta h} \int_h^{T+h} e^{-\beta s} P_s f \, ds.$$

Differentiate at $h = 0$ from the right to obtain

$$u_T \in D(A), \qquad A u_T = \beta u_T + e^{-\beta T} P_T f - f. \tag{15}$$

The tails of both sides converge in supremum norm. Closedness of the graph gives

$$R_\beta f \in D(A), \qquad (\beta - A)R_\beta f = f. \tag{16}$$

For $u \in D(A)$, the orbit derivative from Proposition 3.1 yields

$$\frac{d}{dt}(e^{-\beta t}P_t u) = -e^{-\beta t}P_t(\beta u - Au).$$

Integrate from zero to infinity; the endpoint at infinity vanishes by contraction. Hence $R_\beta(\beta u - Au) = u$ as well. This proves the full-domain inverse formula and the bound

$$\beta\|u\|_\infty \le \|\beta u - Au\|_\infty.$$

No self-adjointness was used. The next calculation shows why it would be inappropriate to insert it.

5 An exact nonsymmetry witness

Under ν_q, the Poisson moments satisfy $\int z^n \, d\nu_q = q^n$ for $n \ge 0$. Use $f(z) = z$ and $g(z) = z^2$, both in the finite Fourier generator domain. Since q and r are real,

$$\begin{aligned}\langle f, Ag\rangle_{L^2(\nu_q)} &= -2q + r(1 - q^2) = 0,\\ \langle Af, g\rangle_{L^2(\nu_q)} &= -q + \tfrac{r}{2}(q^2 - 1) = -2q.\end{aligned}$$

Here the inner product is conjugate linear in its first argument. If $q \neq 0$ the pairings differ. Thus no Hermitian realization agreeing with this core can be inferred from the invariant Poisson law. Invariance, positivity and entropy decay remain valid; they do not require reversibility.

6 Forward densities and a stationary Bregman identity

For an arbitrary probability measure μ, Fubini gives the positive-time density relative to ν_q:

$$f_t(w) = \int_{\mathbb{T}} \frac{p_{\phi_t(z)}(w)}{p_q(w)} \, \mu(dz). \tag{17}$$

The ratio is jointly continuous and strictly positive on the compact product circle for fixed $t > 0$. It has a positive minimum and finite maximum, independent of μ. On every compact time interval inside $(0, \infty)$, the denominators stay uniformly away from zero. Ordinary differentiation of the rational kernel, uniformly in z, w, proves that $t \mapsto f_t$ is continuously differentiable in supremum norm. The same denominator bound gives a bounded angular derivative in w, hence a Lipschitz bound in chord distance on $\mathbb{T}$.

In particular $\int f_t \, d\nu_q = 1$ and $E(t) = \int f_t \log f_t \, d\nu_q$ is finite and differentiable for $t > 0$. Positivity and the uniform bounds on compact time intervals justify the usual scalar derivative,

$$E'(t) = \int_{\mathbb{T}} \dot{f}_t \log f_t \, d\nu_q, \tag{18}$$

since $\int \dot{f}_t \, d\nu_q = 0$. This is a statement about actual forward laws and their Radon–Nikodym density, not an assumed backward-generator domain for $\log f_t$.

For positive a, b, set $B(a,b) = a\log(a/b) - a + b$. It is nonnegative by convexity of $s\log s - s$. If $f\nu_q$ is a probability law with positive continuous f, invariance and the two marginals of $\nu_q(dz)K_h(z,dw)$ give the exact identity

$$\begin{aligned}\int_{\mathbb{T}^2} B(f(z),f(w))\,\nu_q(dz)K_h(z,dw)\\ = \int_{\mathbb{T}} f\log f\,d\nu_q - \int_{\mathbb{T}} \log f\,d((f\nu_q)K_h).\end{aligned} \tag{19}$$

Indeed expansion of B gives four terms. The first integrates to $\int f\log f\,d\nu_q$; the mixed term is the last term in (19); and the two linear terms cancel because both stationary marginals are ν_q. All integrands are bounded, so Fubini is immediate. For $f = f_t$, Chapman–Kolmogorov identifies the last law as μ_{t+h}. Divide by h and use the supremum-norm derivative of f_t to conclude

$$\lim_{h\downarrow 0}\frac{1}{h}\int B(f_t(z),f_t(w))\,\nu_q(dz)K_h(z,dw) = -E'(t). \tag{20}$$

7 The singular jump limit and its domination

Let $v_h(z) = \phi_h(z)$, $z \in \mathbb{T}$. Equation (10) gives

$$\frac{d}{dh}(1-|v_h(z)|^2)\Big|_{h=0} = 2. \tag{21}$$

To check the independence of q, compute $-2\,\mathrm{Re}(\bar{z}V(z))$. The $-z$ term contributes two, while $\bar{z}(1-z^2) = \bar{z} - z$ is purely imaginary. The derivatives are uniform on the compact circle. For $z \neq w$ it follows that

$$\frac{p_{v_h(z)}(w)}{h} \longrightarrow \frac{2}{|w-z|^2}. \tag{22}$$

A pointwise limit alone does not justify the singular integral in (6); we give a bound that also handles points arbitrarily close to the diagonal.

Uniform differentiability and (21) imply that, for all sufficiently small $h > 0$,

$$|v_h(z) - z| \le Ch, \qquad h \le 1 - |v_h(z)|^2 \le 3h,$$

uniformly in z. Since $|w - v_h(z)| \ge 1 - |v_h(z)| \ge h/2$, the triangle inequality gives

$$|w-z| \le |w - v_h(z)| + Ch \le (1+2C)|w - v_h(z)|.$$

Consequently

$$h^{-1}p_{v_h(z)}(w)|w-z|^2 \le 3(1+2C)^2. \tag{23}$$

Suppose $f \ge \delta > 0$ and $|f(z) - f(w)| \le L|w-z|$. Taylor's integral remainder for $s\log s - s$ yields

$$0 \le B(f(z),f(w)) \le \frac{L^2}{2\delta}|w-z|^2.$$

Combine this bound with (23) and the positive lower bound of p_q on the circle. The functions

$$\frac{p_{v_h(z)}(w)}{h\,p_q(w)}B(f(z),f(w))$$

are uniformly bounded on $\mathbb{T}^2$, with value zero when $z = w$. Their pointwise limit is the singular integrand in (6), with the stated zero diagonal convention. Dominated convergence proves its integrability and the equality of its integral with the limit in (20). Apply this to f_t using the positivity and Lipschitz bounds from the preceding section. This proves the exact production formula.

The drift contributes no separate square-gradient term to this formula. Its effect is retained in f_t and the invariant law; (21) shows that the off-diagonal jump rate is the Cauchy rate. No diffusion chain rule or symmetry assumption enters this argument.

8 A nonsharp entropy inequality and decay from initial time

For $w \in \mathbb{T}$,

$$p_q(w) \le \frac{1+|q|}{1-|q|}, \qquad |w-z| \le 2.$$

Thus away from the diagonal $2/(|w-z|^2 p_q(w)) \ge c_q$. Since $B(f(z), f(z)) = 0$, the product with B satisfies the same inequality everywhere under the zero convention. For a normalized positive Lipschitz f bounded away from zero, Fubini gives

$$\begin{aligned}\int_{\mathbb{T}^2} B(f(z), f(w))\, d\nu_q(z) d\nu_q(w) &= \int_{\mathbb{T}} f \log f \, d\nu_q - \int_{\mathbb{T}} \log f \, d\nu_q \\ &\ge \operatorname{Ent}_{\nu_q}(f).\end{aligned}$$

The last step is Jensen's inequality for the logarithm and $\int f \, d\nu_q = 1$. We have proved the modified logarithmic-Sobolev inequality in the exact form

$$c_q \operatorname{Ent}_{\nu_q}(f) \le \int_{\mathbb{T}^2} \frac{2B(f(z), f(w))}{|w-z|^2 p_q(w)} \, \nu_q(dz)\nu_q(dw). \tag{24}$$

This is a jump-Bregman inequality on the displayed positive Lipschitz class, rather than a sharp quadratic Dirichlet-form logarithmic-Sobolev assertion.

For the actual f_t, the production formula implies $E'(t) \le -c_q E(t)$ for $t > 0$. Multiplying by $e^{c_q t}$ and integrating yields

$$E(t) \le e^{-c_q(t-s)} E(s), \qquad 0 < s \le t. \tag{25}$$

To reach initial time, we also need entropy contraction without any initial smoothness.

If $D(\mu\|\nu_q) < \infty$, write $\mu = f\nu_q$. For $h > 0$ put $R_h(z,w) = p_{\phi_h(z)}(w)/p_q(w)$. Invariance says $\int R_h(z,w)\nu_q(dz) = 1$ for ν_q-almost every w; the row integral is also one. The output density is $f_h(w) = \int R_h(z,w) f(z) \nu_q(dz)$. Apply Jensen's inequality pointwise for the nonnegative convex function $\Psi(a) = a\log a - a + 1$, then integrate:

$$\int \Psi(f_h)\, d\nu_q \le \iint R_h(z,w)\Psi(f(z))\, d\nu_q(z) d\nu_q(w) = \int \Psi(f)\, d\nu_q.$$

Tonelli justifies the extended nonnegative integrals. This is $E(h) \le D(\mu\|\nu_q)$. Insert it in (25) and let $s \downarrow 0$ to obtain (7) when the initial entropy is finite. If it is infinite the same extended-value inequality is immediate; at $t = 0$ it is equality. Positive-time finiteness has already been proved directly from (17), even for singular initial laws.

For any fixed $s > 0$, (25) also proves $E(t) \to 0$ for every initial probability law as $t \to \infty$. If a probability measure satisfies $\mu K_h = \mu$ for one $h > 0$, it has finite entropy by positive-time smoothing. Equation (7) then forces $E \le e^{-c_q h} E$, so $E = 0$. Strict convexity of Ψ gives $\mu = \nu_q$. This proves the uniqueness assertion without a path-space construction or a reversibility premise.

9 Classical context and exact verification scope

The Poisson kernel and harmonic-composition construction are classical. The pole equation $\dot{a} = -a + (r/2)(1 - a^2)$ is also the constant-forcing, zero-frequency, unit-noise specialization of the Cauchy-driven phase equation in Tanaka [1], equations (9) and (17). Its relation to a known invariant family must therefore be acknowledged; we do not present the rational pole flow as a new general mechanism. The generator/resolvent arguments are standard contraction semigroup theory. The complete kernel-to-forward-entropy exposition and its precise formal domains are the focus here. Priority for this explicit combined formulation has not been established.

All mapped statements use the frozen verified baseline

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The accompanying theorem map covers the actual Markov kernels, Feller semigroup, closed generator, smoothed core, Laplace resolvent, forward density, stationary Bregman identity, singular limit, modified entropy bound and finite nonsymmetry witness. The human proof groups the same ingredients into harmonic and Banach-space arguments; intermediate regroupings are not claimed as new named Lean theorems.

The results concern the displayed circle kernels and the continuous-function generator. A trajectory law, maximal forward-density generator domain, unsmoothed Fourier graph-core theorem, sharp entropy constant or diffusion curvature estimate is not asserted here. In particular an invariant wrapped Cauchy law cannot justify replacing the actual nonsymmetric operator by a symmetric diffusion.

References

[1] T. Tanaka, Low-dimensional dynamics of phase oscillators driven by Cauchy noise, primary preprint, arXiv:2009.04273. `https://arxiv.org/abs/2009.04273`.